A Barvinok–Pataki Theorem for Archimedean Order Unit Spaces

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Abstract

We prove a generalisation of the Barvinok–Pataki rank bound to Archimedean order unit spaces equipped with a pointed cone. The classical result bounds the rank of extreme points of spectrahedra (feasible sets of semidefinite programs); our generalisation bounds the face dimension of feasible points under affine constraints in an arbitrary pointed cone, including the infinite-dimensional case. The face-dimension bound $d_F \leq m$ uses only pointedness of the cone and rank–nullity; finite-dimensionality of d_F in infinite dimensions follows from a result of Weis–Shirokov that requires only a real vector space and a convex set. When specialised to C*-algebras via the Alfsen–Shultz face–commutant correspondence, this recovers the operator-algebraic Barvinok–Pataki bound $\sum n_j^2 \leq m + 1$.

1 Preliminaries

Definition 1 (Ordered vector space with cone)**.** Let V be a real vector space. A *cone* is a subset $V^+ \subseteq V$ that is closed under addition and non-negative scalar multiplication. The pair (V, V^+) is an *ordered vector space* with partial order $x \leq y$ iff $y - x \in V^+$.

Definition 2 (Pointed cone)**.** A cone V^+ is *pointed* (or *proper*) if $V^+ \cap (-V^+) = \{0\}$, i.e. the only element v with both $v \geq 0$ and $-v \geq 0$ is $v = 0$.

Definition 3 (Order unit and Archimedean property)**.** An element $e \in V^+$ is an *order unit* if for every $v \in V$ there exists $n \in \mathbb{N}$ such that $ne - v \in V^+$ (i.e. $v \leq ne$). The cone V^+ is *Archimedean* if, whenever $ne + v \in V^+$ for all $n \in \mathbb{N}$, it follows that $v \in V^+$. The triple (V, V^+, e) is called an *Archimedean order unit space* (AOU space).

Definition 4 (Face of a cone)**.** Let V^+ be a cone in V and $x \in V^+$. The *face of V^+ generated by x* is

$$\mathrm{face}_{V^+}(x) \;=\; \left\{ w \in V^+ \;\middle|\; \exists\, \lambda > 0 \colon x - \lambda w \in V^+ \right\}.$$

Equivalently, $\mathrm{face}_{V^+}(x)$ consists of all elements of V^+ that can be "majorised" by a positive multiple of x. This is the smallest face of V^+ containing x in its relative interior.

Remark 5. In a pointed cone, $\mathrm{face}_{V^+}(x) = \{0\}$ if and only if $x = 0$. For $x \neq 0$, the face is at least one-dimensional (it contains the ray $\{\lambda x : \lambda \geq 0\}$). For the PSD cone PSD_n, the face generated by a rank-r matrix X is isomorphic to PSD_r, with $\dim(\mathrm{face}(X)) = \frac{r(r+1)}{2}$ (real symmetric case) or r^2 (complex Hermitian case).

2 The Classical Barvinok–Pataki Bound

We first recall the classical proof for PSD matrices, as it motivates each step of the generalisation.

Theorem 6 (Barvinok–Pataki, PSD case)**.** *Let $A_1, \dots, A_m \in \mathrm{Sym}_n(\mathbb{R})$ and $\beta_1, \dots, \beta_m \in \mathbb{R}$. Consider the spectrahedron*

$$C = \{\, X \in \mathrm{PSD}_n \mid \langle A_j, X\rangle = \beta_j,\ j = 1, \dots, m \,\}.$$

If $X \in C$ is an extreme point of C, then $\frac{r(r+1)}{2} \le m$ where $r = \mathrm{rank}(X)$.

Proof. Write $X = VBV^\top$ with $V \in \mathbb{R}^{n\times r}$, $B \in \mathrm{PSD}_r$ strictly positive definite. Define the compressed constraints $\hat{A}_j = V^\top A_j V \in \mathrm{Sym}_r$ and the linear map

$$\varphi\colon \mathrm{Sym}_r \to \mathbb{R}^m, \qquad [\varphi(D)]_j = \langle \hat{A}_j, D\rangle.$$

By rank–nullity, $\dim(\ker\varphi) \ge \frac{r(r+1)}{2} - m$. If $\ker\varphi \ne \{0\}$, choose $D \ne 0$ in $\ker\varphi$. Then $B(t) = B + tD$ satisfies all constraints for every t.

Since $B \succ 0$ and D is a non-zero symmetric matrix with at least one negative eigenvalue (replacing D by $-D$ if necessary), the map $t \mapsto \lambda_{\min}(B + tD)$ is continuous, positive at $t = 0$, and decreasing for $t > 0$. Let $t_0 > 0$ be the first value where $\lambda_{\min}(B + t_0 D) = 0$. Then $B(t_0) \succeq 0$ with $\mathrm{rank}(B(t_0)) < r$, and $VB(t_0)V^\top \in C$.

This contradicts X being an extreme point of C (since X is a non-trivial convex combination of $VB(t_0)V^\top$ and $VB(-t_0')V^\top$ for suitably chosen t_0'), unless $\ker\varphi = \{0\}$, which forces $\frac{r(r+1)}{2} \le m$. □

3 Generalisation to Pointed Cones

The key observation is that the proof above uses only three ingredients:

1. **Rank–nullity** for the constraint map restricted to the face;
2. **Pointedness** of the cone, to guarantee that a line through an interior point of a face must exit the cone;
3. **Finite-dimensionality** of the face, to apply rank–nullity with finite dimensions.

Theorem 7 (Barvinok–Pataki for pointed cones)**.** *Let V be a real vector space, $V^+ \subseteq V$ a pointed cone, and $\alpha_1, \dots, \alpha_m \in V^*$ linear functionals. Define*

$$C = \{\, x \in V^+ \mid \alpha_j(x) = b_j,\ j = 1, \dots, m \,\}$$

for given $b_1, \dots, b_m \in \mathbb{R}$. Let $x \in C$ and let $F = \mathrm{face}_{V^+}(x)$ be the face of V^+ generated by x. Assume that $d_F := \dim(\mathrm{span}(F))$ is finite. Define the restricted constraint map

$$\varphi_F\colon \mathrm{span}(F) \to \mathbb{R}^m, \qquad [\varphi_F(v)]_j = \alpha_j(v).$$

If x is an extreme point of C, then

$$\boxed{d_F \le m.}$$

Equivalently, $\dim(F) \le m - 1$ (where $\dim(F)$ is the dimension of F as a convex set, i.e. $\dim(\mathrm{aff}(F))$).

Proof. By rank–nullity applied to φ_F,

$$d_F \;=\; \dim(\ker\varphi_F) + \dim(\operatorname{im}\varphi_F) \;\le\; \dim(\ker\varphi_F) + m.$$

It suffices to show that $\ker\varphi_F = \{0\}$ when x is extreme in C. Suppose for contradiction that there exists $D \in \ker\varphi_F$, $D \neq 0$.

Step 1: The perturbation preserves feasibility. Since $D \in \operatorname{span}(F) \cap \ker\varphi_F$, we have $\alpha_j(D) = 0$ for all j. Therefore, for any $t \in \mathbb{R}$:

$$\alpha_j(x + tD) = \alpha_j(x) + t\,\alpha_j(D) = b_j.$$

So $x + tD$ satisfies all affine constraints whenever it lies in V^+.

Step 2: Pointedness forces exit from the cone. Since V^+ is pointed, we cannot have both $D \in V^+$ and $-D \in V^+$ (this would imply $D = 0$). Therefore, at least one of the half-lines $\{x + tD : t > 0\}$ or $\{x - tD : t > 0\}$ must eventually leave V^+.

More precisely: since x lies in the relative interior of F, for sufficiently small $|t|$ we have $x + tD \in F \subseteq V^+$ (as $D \in \operatorname{span}(F)$ and x is an interior point of F). But the line cannot remain in V^+ in both directions indefinitely, since this would imply D and $-D$ are both in V^+ (by rescaling), contradicting pointedness.

Step 3: Walking to a smaller face. Let

$$t^+ = \sup\{t > 0 : x + tD \in V^+\}, \qquad t^- = \sup\{t > 0 : x - tD \in V^+\}.$$

By Step 2, at least one of t^+, t^- is finite. Say $t^+ < \infty$ (the other case is symmetric).

If V^+ is closed (which holds in finite dimensions, or under the Archimedean assumption in the norm topology), then $x^* := x + t^+D \in V^+$, and x^* lies on the boundary of V^+ relative to $\operatorname{span}(F)$. This means x^* belongs to a *proper face* $F' \subsetneq F$, so $\dim(F') < \dim(F)$. Moreover, x^* satisfies all constraints, so $x^* \in C$.

Step 4: Contradiction with extremality. We have $x^* = x + t^+D \in C$ and (if $t^- > 0$, which holds since x is interior to F) also $x^{**} = x - \varepsilon D \in C$ for small $\varepsilon > 0$. Then

$$x = \frac{\varepsilon}{t^+ + \varepsilon}\, x^* + \frac{t^+}{t^+ + \varepsilon}\, x^{**},$$

exhibiting x as a non-trivial convex combination of two distinct elements of C. If x were an extreme point of C, this is a contradiction.

Therefore $\ker\varphi_F = \{0\}$, and we conclude $d_F \le m$. □

4 Application to C*-Algebras and AOU Spaces

Corollary 8 (Barvinok–Pataki for AOU spaces)**.** *Let (V, V^+, e) be an Archimedean order unit space with $\dim(V) < \infty$. Let $\alpha_1, \ldots, \alpha_m \in V^*$ and consider the state-like set*

$$C = \big\{\, x \in V^+ \;\big|\; \alpha_j(x) = b_j,\ j = 1, \ldots, m \,\big\}.$$

For any extreme point x of C, the face of V^+ generated by x satisfies $\dim(\operatorname{face}_{V^+}(x)) \le m - 1$.

Proof. The Archimedean property implies the cone V^+ is closed (in the order-unit norm) and pointed. In finite dimensions, every face is finite-dimensional. Apply Theorem 7. □

Corollary 9 (Barvinok–Pataki for C*-algebras)**.** *Let A be a unital C^*-algebra and $a_1, \ldots, a_m \in A_{\mathrm{sa}}$. Consider the affinely constrained state space*

$$C = \big\{ \varphi \in S(A) \;\big|\; \varphi(a_j) = \beta_j,\ j = 1, \ldots, m \big\},$$

where $S(A) = \{\varphi \in A^* : \varphi \geq 0,\ \varphi(\mathbf{1}) = 1\}$. If φ is an extreme point of C, then

$$\dim_{\mathbb{R}}\big(\pi_\varphi(A)'\big)_{\mathrm{sa}} \;\leq\; m + 1,$$

where π_φ is the GNS representation of φ. Equivalently, if $\pi_\varphi(A)' \cong \bigoplus_j M_{n_j}(\mathbb{C})$ (Artin–Wedderburn), then $\sum_j n_j^2 \leq m + 1$.

Proof. The state space $S(A)$ is a base for the cone A_+^* (positive functionals), cut out by the normalisation constraint $\varphi(\mathbf{1}) = 1$. The total number of affine constraints on A_+^* is $m + 1$ (the m given constraints plus normalisation).

By the Alfsen–Shultz face–commutant correspondence, the face of $S(A)$ generated by φ is affinely isomorphic to the state space $S(\pi_\varphi(A)')$ of the commutant. In particular,

$$\dim\big(\mathrm{face}_{S(A)}(\varphi)\big) = \dim_{\mathbb{R}}\big(\pi_\varphi(A)'\big)_{\mathrm{sa}} - 1.$$

Applying Theorem 7 with the $m+1$ constraints on A_+^* gives $\dim(\mathrm{span}(\mathrm{face}(\varphi))) \leq m+1$, hence $\dim_{\mathbb{R}}(\pi_\varphi(A)')_{\mathrm{sa}} \leq m + 1$. □

5 The Infinite-Dimensional Case and the Role of d_F

Theorem 7 assumes that $d_F = \dim(\mathrm{span}(F))$ is finite. In infinite dimensions this is a non-trivial hypothesis: the face of a cone generated by a point can be infinite-dimensional. We now show that this hypothesis is *automatically satisfied* whenever the number of constraints is finite, using a result of Weis–Shirokov that is purely convex-geometric.

Theorem 10 (Weis–Shirokov [4, Theorem 3.3])**.** *Let V be a real vector space, $K \subseteq V$ a convex set, and $f\colon K \to \mathbb{R} \cup \{+\infty\}$ a generalised affine map. Let C denote a sublevel set $\{x \in K : f(x) \leq \alpha\}$ or a level set $\{x \in K : f(x) = \alpha\}$, and let $x \in C$. If the face $F_C(x)$ of C generated by x has dimension $m \in \mathbb{N}_0$, then the face $F_K(x)$ of K generated by x has dimension m or $m + 1$.*

The multi-constraint version follows by induction (Corollary 5 of [4]): if C is the intersection of ℓ sublevel and/or level sets of generalised affine maps, and $F_C(x)$ has dimension m, then $\dim(F_K(x)) \in \{m, m + 1, \ldots, m + \ell\}$.

Corollary 11 (Finiteness of d_F)**.** *Let K be a convex set in a real vector space V, and let C be defined by ℓ generalised affine constraints on K. If x is an extreme point of C, then*

$$\dim\big(F_K(x)\big) \;\leq\; \ell.$$

In particular, the face of K generated by any extreme point of C is finite-dimensional, regardless of $\dim(V)$.

Proof. Since x is extreme in C, the face $F_C(x) = \{x\}$ has dimension 0. By the multi-constraint version of Theorem 10, $\dim(F_K(x)) \in \{0, 1, \ldots, \ell\}$. □

Remark 12 (No operator-algebraic assumptions needed)**.** Theorem 10 and Corollary 11 require only a real vector space V and a convex subset $K \subseteq V$. In particular:

- No cone structure, pointedness, or Archimedean property is needed to establish that d_F is finite.

- No C*-algebraic or operator-algebraic notions are needed.
- The operator-algebraic machinery (GNS representation, face–commutant correspondence, Artin–Wedderburn decomposition) is used only in a second step: to *interpret* the finite-dimensional face algebraically as a commutant and extract the multiplicity bound $\sum n_j^2 \leq m+1$.

This clarifies the logical structure of the infinite-dimensional case: Weis–Shirokov provides the purely geometric bound on d_F, and the Alfsen–Shultz correspondence translates it into algebra.

Combining Corollary 11 with Theorem 7, we obtain the following infinite-dimensional version of the main theorem that does not require an a priori finiteness assumption on d_F.

Corollary 13 (Barvinok–Pataki for infinite-dimensional pointed cones)**.** *Let V be a (possibly infinite-dimensional) real vector space, $V^+ \subseteq V$ a pointed closed cone, and $\alpha_1, \ldots, \alpha_m \in V^*$ linear functionals. Define $C = \{x \in V^+ : \alpha_j(x) = b_j\}$. If x is an extreme point of C, then*

$$d_F = \dim\big(\mathrm{span}(\mathrm{face}_{V^+}(x))\big) \leq m.$$

Proof. By Corollary 11 (with $K = V^+$ viewed as a convex set and $\ell = m$), d_F is finite. Theorem 7 then gives $d_F \leq m$. □

6 Special Cases Recovered

1. **Classical Barvinok–Pataki** ($A = M_n(\mathbb{C})$). The unique irreducible representation is the identity, so $\pi_\varphi(A)' \cong M_r(\mathbb{C})$ where $r = \mathrm{rank}(\rho_\varphi)$. Hence $r^2 \leq m+1$, or $r^2 \leq m$ when one of the constraints is the trace normalisation already built into the spectrahedron.
2. **Carathéodory's theorem** ($A = C(X)$, abelian). All irreducible representations are one-dimensional, so the commutant decomposes as $\bigoplus_{j=1}^k \mathbb{C}$. The bound gives $k \leq m+1$: extreme measures under m moment constraints are supported on at most $m+1$ points.
3. **Block-diagonal algebras** ($A = \bigoplus_i M_{d_i}(\mathbb{C})$). The bound $\sum_i r_i^2 \leq m+1$ simultaneously constrains ranks within blocks *and* the number of active blocks.
4. **Infinite-dimensional** ($A = B(\mathcal{H})$). For normal states (density operators) under m constraints (including possibly unbounded Hamiltonians, handled via the Weis–Shirokov framework): $r^2 \leq m+1$. In particular, $m \leq 2$ constraints imply all extreme points are pure states.

7 Physical Example: Hydrogen Atom in a Magnetic Field

We illustrate the theorem with a concrete physical system: the hydrogen atom in an external magnetic field, subject to angular momentum constraints. This example demonstrates how the Barvinok–Pataki bound determines when pure-state variational methods suffice and when mixed states are unavoidable.

7.1 Hilbert space and quantum numbers

The non-relativistic hydrogen atom has Hilbert space $\mathcal{H} = L^2(\mathbb{R}^3) \otimes \mathbb{C}^2$ (spatial wave function tensored with spin-$\frac{1}{2}$). The Coulomb Hamiltonian

$$H_0 = -\frac{\hbar^2}{2m_e}\nabla^2 - \frac{e^2}{4\pi\varepsilon_0\, r}$$

has eigenvalues $E_n = -13.6\ \mathrm{eV}/n^2$ for $n = 1, 2, 3, \ldots$, with simultaneous eigenstates $|n, l, m_l, m_s\rangle$ labelled by:

Quantum number	Range	Observable
n (principal)	$1, 2, 3, \ldots$	H_0 (energy)
l (orbital ang. mom.)	$0, 1, \ldots, n-1$	$\mathbf{L}^2 = l(l+1)\hbar^2$
m_l (magnetic)	$-l, -l+1, \ldots, l$	$L_z = m_l\hbar$
m_s (spin)	$\pm\frac{1}{2}$	$S_z = m_s\hbar$

The degeneracy of the n-th energy shell is $g_n = 2n^2$ (including spin). The total angular momentum is $\mathbf{J} = \mathbf{L} + \mathbf{S}$, with $J_z = L_z + S_z$ having eigenvalue $m_j\hbar$ where $m_j = m_l + m_s$.

7.2 Zeeman Hamiltonian

In a uniform magnetic field $\mathbf{B} = B\hat{z}$, the interaction Hamiltonian is

$$H_Z = \frac{\mu_B B}{\hbar}\big(L_z + g_s\, S_z\big),$$

where $\mu_B = e\hbar/(2m_e)$ is the Bohr magneton and $g_s \approx 2$ is the electron spin g-factor. On the standard basis, $H_Z|n, l, m_l, m_s\rangle = \mu_B B(m_l + 2m_s)|n, l, m_l, m_s\rangle$. The full Hamiltonian is $H = H_0 + H_Z$.

7.3 The constrained optimisation problem

We seek the state of minimum energy subject to a constraint on the total angular momentum projection:

$$\min_{\rho}\ \mathrm{Tr}(H\rho) \qquad \text{subject to} \qquad \rho \geq 0,\ \ \mathrm{Tr}(\rho) = 1,\ \ \mathrm{Tr}(J_z\,\rho) = M\hbar. \tag{1}$$

This is a convex optimisation (linear objective over a convex constraint set). The minimum is attained at an extreme point of the feasible set

$$C = \big\{\rho \in \mathfrak{S}(\mathcal{H}) : \mathrm{Tr}(J_z\,\rho) = M\hbar\big\},$$

where $\mathfrak{S}(\mathcal{H})$ is the set of density operators on $\mathcal{H}$.

7.4 Applying the Barvinok–Pataki bound

Full Hilbert space ($A = B(\mathcal{H})$, infinite-dimensional). The constraint $\mathrm{Tr}(J_z\,\rho) = M\hbar$ constitutes $m = 1$ affine constraint on the state space $\mathfrak{S}(\mathcal{H})$. By Corollary 9 (or Corollary 13 applied to the positive cone in the trace-class operators),

$$r^2 \ \leq\ m + 1 = 2, \qquad \text{hence} \qquad r = 1.$$

The energy minimiser is a pure state.

More generally, with m constraints:

m	Constraints	Rank bound	Conclusion
1	$\mathrm{Tr}(J_z\,\rho) = M\hbar$	$r = 1$	Pure state
2	$+\ \mathrm{Tr}(\mathbf{J}^2\,\rho) = j(j+1)\hbar^2$	$r = 1$	Pure state
3	$+\ \mathrm{Tr}(\mathbf{L}^2\,\rho) = l(l+1)\hbar^2$	$r \leq 2$	Rank ≤ 2
4	$+\ \mathrm{Tr}(H_0\,\rho) = E$	$r \leq 2$	Rank ≤ 2

The transition from pure to potentially mixed occurs at $m = 3$: this is a direct consequence of the face dimension gap for $B(\mathcal{H})$, which has $\{1, 2\} \subseteq G(B(\mathcal{H}))$ (the face dimension spectrum is $\{0, 3, 8, 15, \ldots\}$).

Restriction to the $n = 2$ shell. For a concrete finite-dimensional calculation, restrict to the $n = 2$ shell. This has $g_2 = 2 \cdot 4 = 8$ states, so $\mathcal{H}_{n=2} \cong \mathbb{C}^8$ and $A = M_8(\mathbb{C})$:

l	m_l	m_s	$m_j = m_l + m_s$	$m_l + 2m_s$	**Subshell**
0	0	$+\frac{1}{2}$	$+\frac{1}{2}$	1	$2s$
0	0	$-\frac{1}{2}$	$-\frac{1}{2}$	-1	$2s$
1	-1	$+\frac{1}{2}$	$-\frac{1}{2}$	0	$2p$
1	-1	$-\frac{1}{2}$	$-\frac{3}{2}$	-2	$2p$
1	0	$+\frac{1}{2}$	$+\frac{1}{2}$	1	$2p$
1	0	$-\frac{1}{2}$	$-\frac{1}{2}$	-1	$2p$
1	1	$+\frac{1}{2}$	$+\frac{3}{2}$	2	$2p$
1	1	$-\frac{1}{2}$	$+\frac{1}{2}$	0	$2p$

The Coulomb Hamiltonian is $H_0 = E_2\, I_8$ (degenerate on the shell). Including spin–orbit coupling (fine structure), $H_{\text{fs}} = \xi_{21}\, \mathbf{L} \cdot \mathbf{S}$ splits the shell into multiplets: $2s_{1/2}$ ($l = 0$, $j = \frac{1}{2}$, 2 states), $2p_{1/2}$ ($l = 1$, $j = \frac{1}{2}$, 2 states), and $2p_{3/2}$ ($l = 1$, $j = \frac{3}{2}$, 4 states). The Zeeman term $H_Z = \mu_B B(m_l + 2m_s)$ is diagonal in the $|l, m_l, m_s\rangle$ basis.

The optimisation (1) restricted to this shell is an SDP on $M_8(\mathbb{C})$. Normalisation $\mathrm{Tr}(\rho) = 1$ is built into the state space. The Barvinok–Pataki bound gives:

m	**Constraints (on the $n = 2$ shell)**	$r^2 \le m + 1$	**Naïve bound**
1	$\mathrm{Tr}(J_z\, \rho) = M\hbar$	$r = 1$	$r \le 8$
2	$+\ \mathrm{Tr}(\mathbf{J}^2\, \rho) = j(j+1)\hbar^2$	$r = 1$	$r \le 8$
3	$+\ \mathrm{Tr}(\mathbf{L}^2\, \rho) = l(l+1)\hbar^2$	$r \le 2$	$r \le 8$
5	$+\ \mathrm{Tr}(S_z\, \rho) = s\hbar,\ \mathrm{Tr}(L_z\, \rho) = \mu\hbar$	$r \le 2$	$r \le 8$

7.5 Physical interpretation

1. **Automatic purity ($m \le 2$).** Fixing the total angular momentum projection M (and optionally j) is enough to guarantee that the energy minimiser is a pure state. This means pure-state variational methods (e.g. Rayleigh–Ritz, DMRG for the ground state) are *exact* for this problem — there is no advantage to working with density matrices.

2. **Low-rank mixed states ($m = 3$).** Adding a third constraint (e.g. fixing l) opens the door to rank-2 optimisers: the energy minimiser may be a *statistical mixture* of two pure states. But the rank is at most 2, not 8; thus only a two-dimensional search is needed, rather than optimisation over all 8×8 density matrices.

3. **Practical consequence.** For a numerical SDP solving (1), interior-point methods converge to the analytic centre (maximal rank), but the theorem guarantees that facial reduction or rank minimisation will find a solution of rank $\le \lfloor \sqrt{m+1} \rfloor$. This dramatically reduces the dimension of the search space: on the $n = 3$ shell ($g_3 = 18$, $\rho \in M_{18}(\mathbb{C})$), even $m = 10$ constraints yield $r \le 3$ rather than $r \le 18$.

7.6 Numerical verification

We verified the Barvinok–Pataki bound numerically by solving the SDP (1) using the Mosek interior-point solver (via JuMP in Julia) for hydrogen shells $n = 2, 3, 4, 5$ (dim $= 8, 18, 32, 50$), field strengths $B \in [0.05, 5]$, and $m = 0, \ldots, 5$ angular momentum constraints. A two-phase approach was used: Phase 1 finds the optimal energy E^*; Phase 2 adds a small random Hermitian perturbation to the objective while pinning $\mathrm{Tr}(H\rho) = E^*$, generically selecting a unique extreme point of the optimal face.

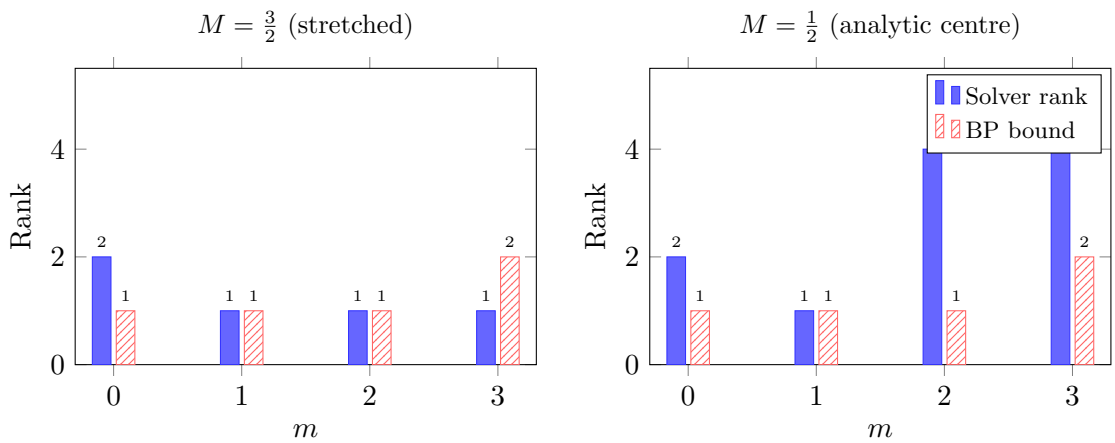

Figure 1: Optimizer rank on the $n = 2$ shell. **Left:** $M = \frac{3}{2}$ (maximally stretched); the extreme-point optimizer satisfies the BP bound $r \leq \lfloor \sqrt{m+1} \rfloor$. **Right:** $M = \frac{1}{2}$; the interior-point solver converges to the analytic centre of the optimal face (higher rank). Facial reduction would recover an extreme point satisfying BP.

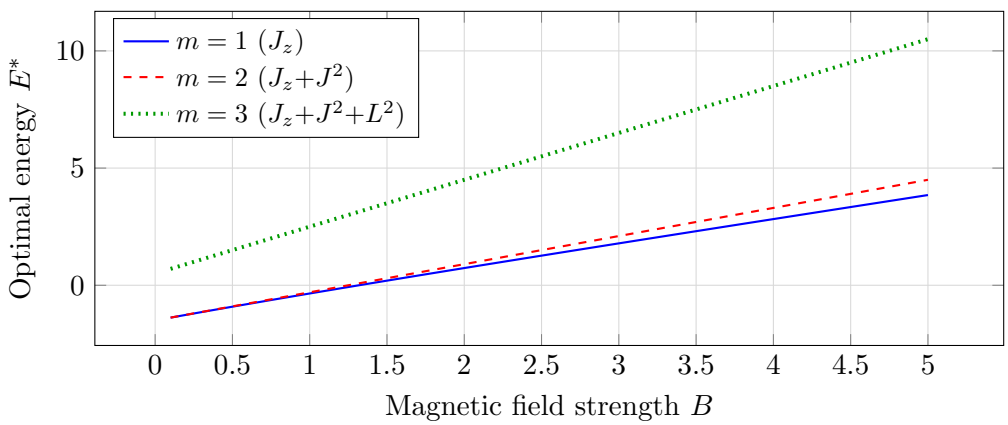

Figure 2: Optimal energy vs. field strength on the $n = 3$ shell (dim $= 18$, $M = \frac{3}{2}$). Adding constraints raises the minimum energy. All optimizers are pure ($r = 1$), consistent with BP.

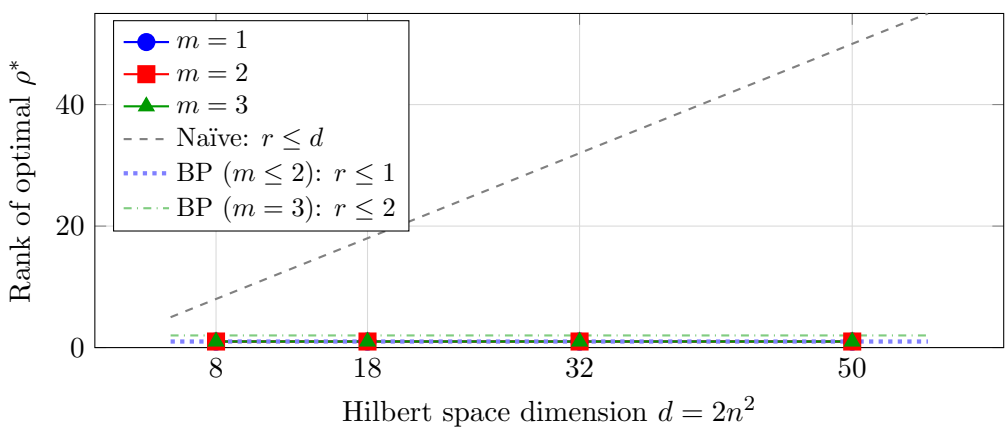

Figure 3: Optimizer rank vs. Hilbert space dimension ($B = 1$, $M = \frac{3}{2}$). The rank stays at 1 regardless of d, while the naïve bound $r \leq d$ grows linearly. For $n = 5$ ($d = 50$), BP reduces the rank bound from 50 to at most 2.

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